

ZHX2 crop comparison (MoE-base): the zinc-finger signal is erased when homeodomains are added

1. ZF1+ZF2 only | raw 20-col (GC 0.61)



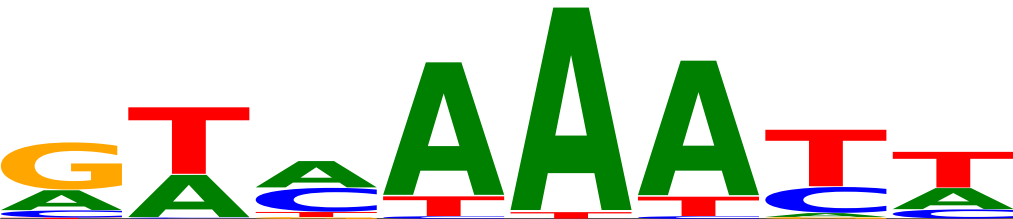
1. ZF1+ZF2 only | gate core GTCTCAAGGGGGG GC 0.59 LC 0.54 r=0.51



2. HD2+HD3 (primary) | raw 20-col (GC 0.18)



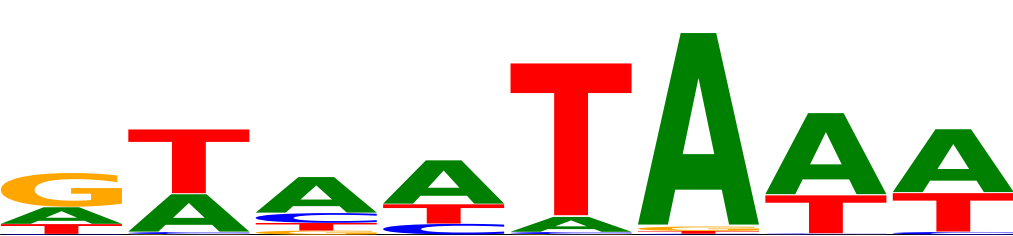
2. HD2+HD3 (primary) | gate core GTAAATTT GC 0.21 LC 0.50 r=0.26



3. ZF + HD2+HD3 | raw 20-col (GC 0.11)



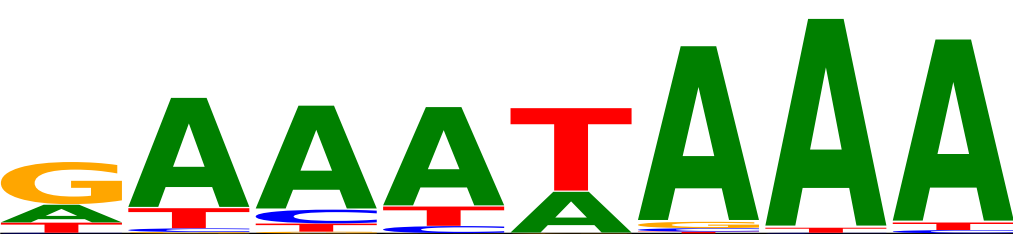
3. ZF + HD2+HD3 | gate core GTAATAAA GC 0.13 LC 0.62 r=0.14



4. all-DBD (6 domains) | raw 20-col (GC 0.10)



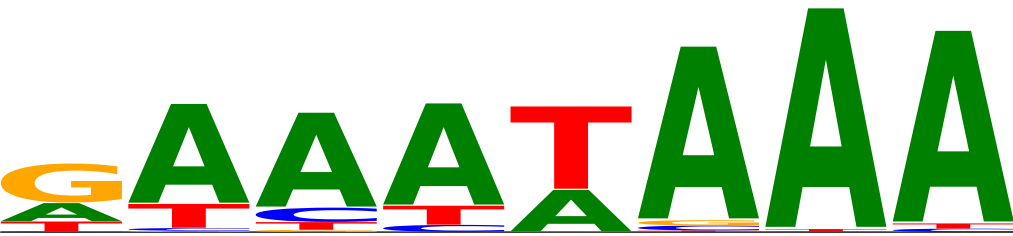
4. all-DBD (6 domains) | gate core GAAATAAA GC 0.11 LC 0.75 r=0.50



5. full sequence | raw 20-col (GC 0.10)



5. full sequence | gate core GAAATAAA GC 0.11 LC 0.75 r=0.48



reference AGGCTGAG (Zhang 2023 Nat Commun; ChIP-seq + EMSA)



red = AT-rich prediction (GC<0.35); homeodomain prior
blue = GC-matched to reference

Domain swamping: pooled DBD mean is dominated by the largest masked domain. Only the ZF-only crop recovers the reference's GC composition.